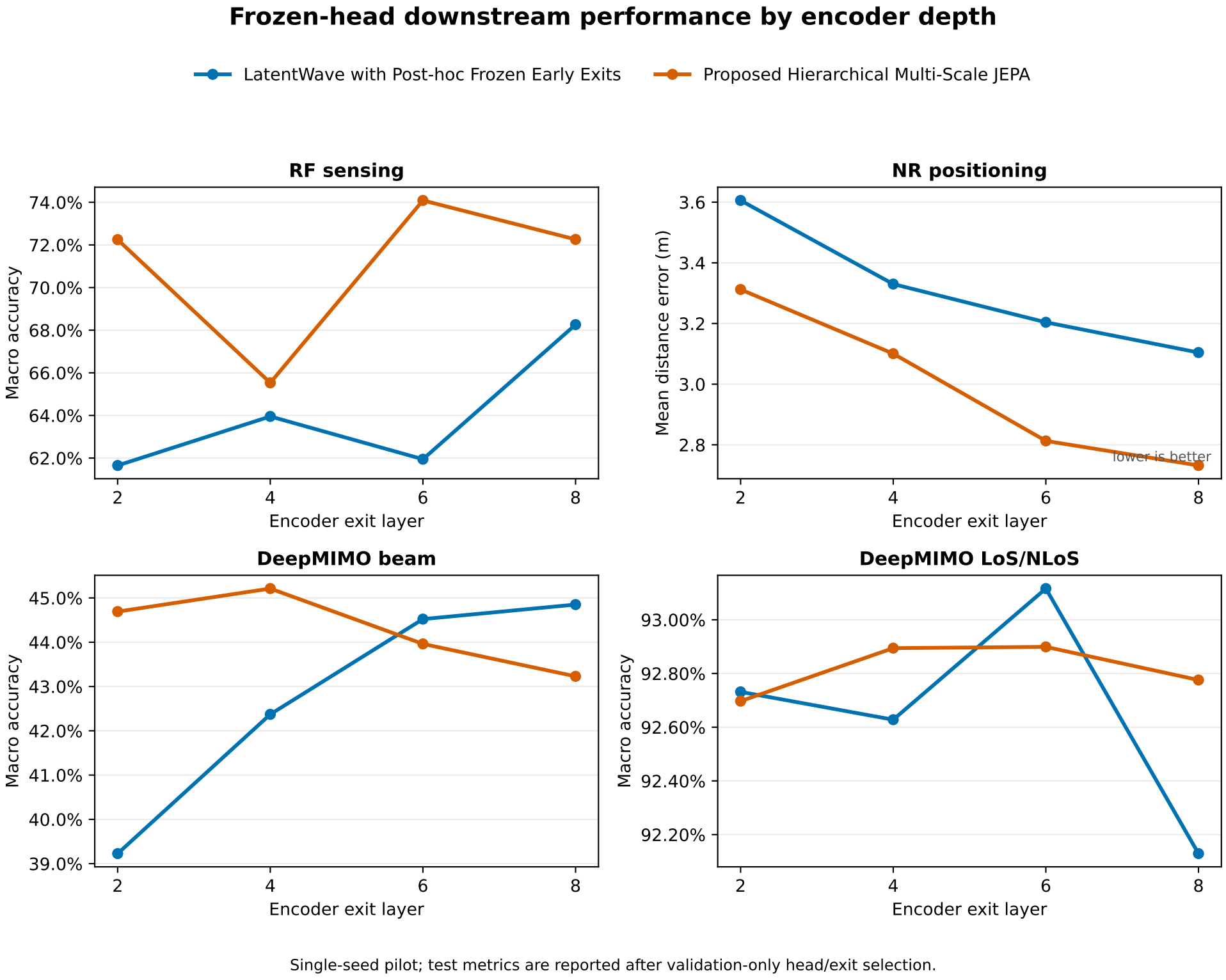
Frozen-head downstream performance by encoder depth
LatentWave with Post-hoc Frozen Early Exits
Proposed Hierarchical Multi-Scale JEPA
RF sensing
Macro accuracy
74.0%
72.0%
70.0%
68.0%
66.0%
64.0%
62.0%
2
4
6
8
Encoder exit layer
NR positioning
Mean distance error (m)
3.6
3.4
3.2
3.0
2.8
lower is better
Encoder exit layer
DeepMIMO beam
Macro accuracy
45.0%
44.0%
43.0%
42.0%
41.0%
40.0%
39.0%
Encoder exit layer
DeepMIMO LoS/NLoS
Macro accuracy
93.00%
92.80%
92.60%
92.40%
92.20%
Encoder exit layer
Single-seed pilot; test metrics are reported after validation-only head/exit selection.